

Wing-Spar Material Trade Study

Compare candidate materials for a model wing spar and recommend the best option for the mission.

Intermediate	1-2 class periods	Aerospace Mission
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Engineering Context

A small unmanned aircraft needs a wing spar that is stiff, light, affordable, and manufacturable with available tools. No single material is best in every category.

What You Will Practice

- Compare material properties
- Create weighted decision criteria
- Conduct simple bending tests
- Defend a materials recommendation

What You Need

- Samples of wood, plastic, aluminum, composite substitute, or approved materials
- Mass scale
- Ruler
- Test fixture
- Small loads
- Decision matrix

Student Instructions

- 1 Define mission priorities for mass, stiffness, cost, availability, and manufacturability.
- 2 Measure sample dimensions and mass.
- 3 Perform a consistent bending test on each material using the same span and load.
- 4 Record deflection and any permanent damage.
- 5 Normalize or rank each material for the five criteria.
- 6 Create a weighted decision matrix.
- 7 Recommend one material and identify the biggest risk in that choice.

Engineering Requirements

- At least four candidate materials
- Same span, support spacing, and test load for every sample
- Decision matrix must include at least five criteria
- Recommendation must cite both test data and mission priorities

Constraints & Safety

- Stay below instructor-approved load limits
- Wear eye protection during bending tests
- Do not test brittle materials near faces
- Handle sharp metal edges carefully

What You Submit

- Material property table
- Bending-test data
- Weighted decision matrix
- Mission recommendation
- Risk statement

Reflection Questions

- 1 Which criterion most strongly affected the result?
- 2 Did the stiffest material have the best overall score?
- 3 How would production quantity change the decision?
- 4 What missing property would you test next?

Engineering Tip Compare materials using equivalent geometry whenever possible; geometry can dominate stiffness.	Common Mistake A decision matrix is not objective if weights are chosen after seeing the test results.
Stretch Goal Repeat the trade study for a different mission that prioritizes cost or impact resistance.	Career Connection Materials and structures engineers select alloys, polymers, composites, and sandwich structures based on mission-specific tradeoffs.